Project 2 on Neural networks

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This project ...

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I. INTRODUCTION

II. THEORY

A. Cost functions and their gradients

The loss function is the most important function for training, since training a model means to minimize the cost function of the model. To minimize the cost function, we use backpropagation to find the gradient of the cost, with regards to and gradient descent.

1. MSE

When we have our calculated output value $\tilde{y}$ and the true output value y, with n data points, the mean-squared error and its derivative is

$$MSE(y,\tilde{y}) = \frac{1}{n}\sum_{i=0}^{n-1}(y-\tilde{y})^2 \quad \frac{1}{n}||y - X\Theta||_2^2 \tag{1}$$

L2 norm which corresponds to the Ridge cost function.

$$MSE + \lambda||\Theta||_2^2 \quad \Theta(W,b) \tag{2}$$

L1 norm which corresponds to the Lasso cost function.

$$MSE + \lambda||\Theta|| \quad \Theta(W,b) \tag{3}$$

2. Binary cross entropy: log loss

For binary classification problems with and without the L_1 and L_2 norms.

$$C(\Theta) = -\sum_{i=1}^{n}(y_i \log p_i + (1-y_i)\log[1-p_i]) \tag{4}$$

$$p_i = \sigma(\theta_0 + \sum_{i}^{n}(\theta_i x_i)) \tag{5}$$

The cross entropy L1 and L2 norm is supplemented like the ordinary MSE. (Hjorth-Jensen, 2016, p. (151-154))

3. Multiclass cross entropy cost function

aka Softmaxx cross entropy or softmax loss function. K classes.

$$C(\Theta) = -\sum_{i=1}^{n}\sum_{k=0}^{K-1} y_{ik}\log p(C=k|x_i) \tag{6}$$

$$p(C=k \mid x) = \frac{\exp(x^T\theta_k)}{\sum_{l=0}^{K-1}\exp(x^T\theta_l)} \quad k = 0,\ldots,K-1 \tag{7}$$

(Hjorth-Jensen, 2016, p. 156)

B. Activation functions and their gradients

1. Sigmoid logit function

$$\sigma(z) = \frac{1}{1+\exp(-z)} \tag{8}$$

$$\sigma'(z) = \sigma(z)(1-\sigma(z)) \tag{9}$$

2. RELU function

$$ReLU(z) = \max(0,z) \tag{10}$$

$$ReLU'(z) = \begin{cases} 0 & z<0 \\ 1 & z>0 \end{cases} \tag{11}$$

3. leaky RELU function

$$leakyReLU(z) = \begin{cases} 0.01z & z<0 \\ z & z>0 \end{cases} \tag{12}$$

$$leakyReLU'(z) = \begin{cases} 0.01 & z<0 \\ 1 & z>0 \end{cases} \tag{13}$$

III. METHODS

A. NeuralNetwork

1. Feed-forward pass

2. Backpropagation

B. Comparison with OLS

C. Activation, depth and width

1. Ridge

2. Lasso

D. Classification

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION

REFERENCES

M. Hjorth-Jensen (2016), *Machine Learning and the Physical Sciences* (Springer).